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Advanced Computer Architecture

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Name
Last Name

The student has to choose 2 out of the 3 exercises. If all the 3 exercises will be completed, just the 2 with the lowest score will be considered. This is an open book exam, which means the students can use all their notes (we do not care of the format) but they cannot share their exams with anyone during the test.

At the end of the exam, when the professor will call it, each student has to upload at most 2 PDF files using the Microsoft Form share via the Zoom chat at the beginning of the exam: NOTE: if you are doing problem 1 and 3, upload your files using the corresponding box 1 and 3.

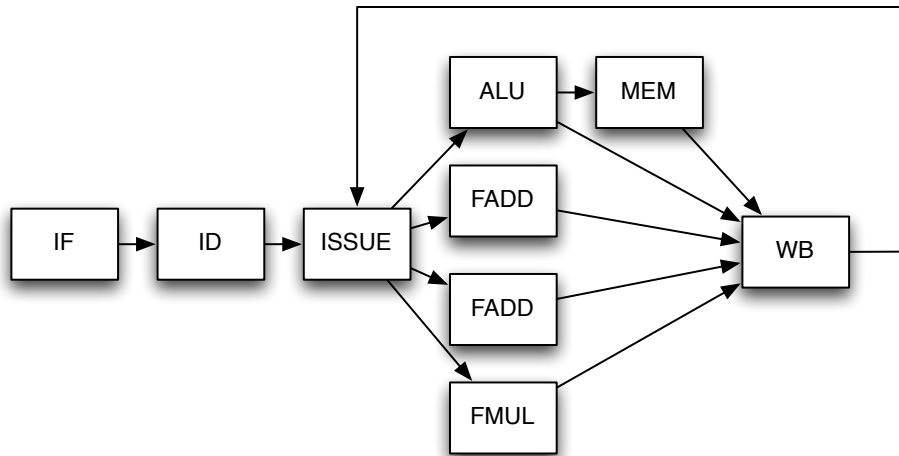
Few notes on the files:

1. You have to submit 1 PDF per question
2. Each PDF has to be max 10MB
3. Each PDF has to be named as codice persona plus the exe, e.g., 10000000_exe1.pdf
4. Each PDF has to report the name and the codice persone on each page
5. All these points are in logic AND, if one of them will not be respected, the ENTIRE exam will not be evaluated

Problem 1 (50%)	
Problem 2 (50%)	
Problem 3 (50%)	
Total (100%)	

Problem 1

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- Memory operations take 2 cycles (includes time in ALU)
- Floating-point add instructions take 3 cycles
- Floating-point multiply instructions take 4 cycles
- There is no register renaming
- Instructions are fetched, decoded and issued in order
- The issue stage is a buffer of unlimited length that holds instructions waiting to start execution
- An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

To simplify book-keeping for this problem we will only track instructions during a few critical stages:

- I - When an instruction enters the issue stage
- E - When an instruction starts execution (enters FU)
- C - When an instruction completes execution (enters WB, results available this cycle)

For this problem we will be using the following code:

```

I1    L.D   F0, 0(R0)
I2    MUL.D F0, F1, F1
I3    ADD.D F2, F1, F1
I4    ADD.D F4, F3, F3
I5    ADDI  R1, R3, 8
I6    ADD.D F6, F0, F0

```

Question1 - Fill in the following table to indicate how the given code will execute. Assume all register values are available at the start of execution.

Cycle Inst	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
I1	I	E		C																
I2				I	E				C											
I3					I	E				C										
I4						I	E				C									
I5							I	E				C								
I6								I	E				C							

Problem 2

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	5	6	7	8
MULTD F0 F4 F2	6	9	19	20
ADD F2 F8 F6	8	9	10	11
DIVD F12 F0 F6	7	21	31	32
SUBD F6 F8 F2	9	10	11	12

Question A. Is there a “configuration” that can respect the shown execution?
If the previous answer was positive, how many units? Which kind? What latency?

Question B. If the previous table was not correct, please, write the right one
and specify the number, kind and latency for each unit.

Answer. A

No, there is no “configuration” that can respect the shown execution

Answer. B

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	5	6	7	8
MULTD F0 F4 F2	6	9	19	20
ADDD F2 F8 F6	9	10	11	12
DIVD F12 F0 F6	10	21	31	32
SUBD F6 F8 F2	11	13	14	22

1 MEM 1 CC
1 MULT 10 CC
1 DIV 10 CC
1 INT 1 CC

That is what we were expecting, but we accepted other solutions but when all the FUs were set to 1 CC, this is non simplification, or “up to the students”, we explained several times that MULT and DIV are more complex operations wrt and ADDD/SUBD, therefore they cannot have the same execution time.

Problem 3

The Cell and Xenon processors are both examples of MIMD.
Please effectively support your answer!

Note: We are effectively supporting the answer, this can be found in the slides/videos.

The Cell and Xenon processors are both examples of MIMD.

The Cell is an heterogeneous multicore where the 8 SPE are SIMD architectures, while the Xenon is an homogenous multicore where each core has been extended wrt the original version of the base PowerCore unit with SIMD extensions.